Chapter 1 Division Algebra

Reference

- A First Course in Noncommutative Rings, T.Y.Lam,Chapter 5;
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1.1 Finite Dimensional Division Algebra

1.1.1 Discussion of Complex Numbers,Quaternion and Frobenius's Theorem

Introduction We have the extension of field

$$\mathbb{R} \rightsquigarrow \mathbb{C} \qquad dim_{\mathbb{R}}(\mathbb{C}) = 2$$

Morever, we want to find some $\mathbb{R}$-algebras, and be able to devide nubmers.

Definition 1.1.1.1 A unital algebra D in which every nonzero element is invertible is called a division algebra.

i.e. for every pair $a \in D - \{0\}$, such x, y exists.

$$ax = 1 \qquad ya = 1$$

A question is:

Is it possible to define multiplication on an n-dimensional real space so that it becomes a real division algebra?

Standard Results:

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Assume: D is an n-dimensional $\mathbb{R}$-algebra.

♢ **Lemma 1.1.1.2** For every $x \in D$, there exists a $\beta \in \mathbb{R}$ such that

$$x^2 + \beta x \in \mathbb{R}$$

◦ **Proof** ... □

Define:

$$V := \{v \in D | v^2 \leq 0\}$$

♢ **Lemma 1.1.1.3** If $u, v \in V$ are linearly independent, then so are u, v and 1.

◦ **Proof** ... □

Jordan Product

Definition 1.1.1.4 (Jordan Product) For $x, y \in D$, we write

$$x \circ y := xy + yx$$

Frobenius's Theorem

▶ ***Theorem*** (Frobenius's Theorem) A **finite dimensional division algebra** D is isomorphic to $\mathbb{R}, \mathbb{C}$ or $\mathbb{H}$.

1.2 Structure of Finite Dimensional Algebras